



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test 2

Semester: III

Programme: B. Tech Information Technology

Course: Data Structure

Time: 1 Hours 30 minutes

Max Marks: 30 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32301	Course Outcome
CO1	To gain knowledge about basic concepts of data structures.
CO2	To acquire knowledge of stacks and queues with their applications.
CO3	To aware about the concepts of trees with their applications.
CO4	To learn and design the algorithm for graphs with their applications.
CO5	To implement basic operations on linked list.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Stack Overflow .	CO3	2	2
b.	What is Adjacency Matrix ?	CO4	2	2
c.	Define Binary Tree .	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Stack operations (Push & Pop) .	CO3	2	4
b.	Explain Applications of Stack .	CO3	4	4

c.	Differentiate between Linear Queue and Circular Queue .	C03	3	4
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Graph representation using Adjacency List .	CO4	2	4
b.	Explain DFS traversal with example .	CO4	3	4
c.	Explain Prim's Algorithm .	CO4	3	4
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Binary Tree Traversals .	CO5	2	4
b.	Describe Binary Search Tree (BST) with operations.	CO5	3	4
c.	Explain AVL Tree and its advantages .	CO5	3	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create